

Rheinwerk MSC-CP200 Centrifugal Process Pump - Operation & Maintenance Manual

RW-CP200-OM-Rev4 | Rheinwerk Pumps GmbH

This manual covers the installation, operation and maintenance of the Rheinwerk MSC-CP200 single-stage overhung centrifugal process pump for hot organic duty. Read Section 2 (Safety) in full before any work. Retain this manual with the equipment records for the life of the pump.

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1. General Description

The MSC-CP200 is a single-stage, overhung, radially-split centrifugal process pump conforming to the general arrangement of ISO 2858 and the API 610 OH2 configuration. It is designed for the transfer of hot organic liquids at temperatures up to 220 C and differential heads up to 160 m. The pump is foot-mounted on a fabricated baseplate and is close-coupled to a standard IEC flange motor through a spacer coupling.

The wetted parts are supplied in carbon steel with 12% chrome internals as standard, with optional stainless and duplex trims for corrosive service. The pump uses a cartridge mechanical seal as standard, arranged for an API Plan 11 flush taken from pump discharge through a flow-control orifice back to the seal chamber.

1.1 Intended Use

The pump is intended for clean to mildly contaminated hot organic liquids within the performance envelope stamped on the nameplate. Operation outside the rated flow, head, temperature or NPSH margin voids the warranty and will shorten seal and bearing life.

Typical services include column bottoms, hot ester feed and product transfer, and hot resin blend transfer. The pump is not suitable for slurries, abrasive service or liquids containing significant solids without a purpose-selected trim and seal plan.

1.2 Nameplate and Identification

Each pump carries a stainless nameplate stating model, serial number, rated duty point, maximum working pressure and test pressure. Always quote the serial number when ordering spares.

1.3 Construction

The pump is of back pull-out construction: the bearing housing, shaft, impeller and seal can be withdrawn from the rear of the casing without disturbing the casing or the connected pipework. This shortens seal and bearing maintenance and is the basis of the cartridge seal replacement procedure in Section 5.3.

The casing is centreline-supported for hot service so that thermal growth is symmetrical about the shaft, preserving alignment as the pump heats to operating temperature. The impeller is keyed and threaded to the shaft and locked against the operating direction of rotation.

1.4 Performance Range

The MSC-CP200 frame covers flows to 220 m³/h and differential heads to 160 m at the maximum trim. Each pump is supplied trimmed to its specific duty; the as-built performance curve is issued with the pump data sheet. Prolonged operation far from the best efficiency point increases radial load, vibration and seal and bearing wear.

2. Safety

Hot organic liquids present burn, fire and toxic-exposure hazards. Before any maintenance the pump must be isolated, de-pressurised, drained, cooled below 45 C and locked out in accordance with the site LOTO procedure. Confirm zero energy before breaking any joint.

Never run the pump dry or against a closed discharge for more than a few seconds. Never operate with guards removed. Personal protective equipment appropriate to the pumped fluid must be worn at all times.

2.1 Residual Hazards

Even after isolation, trapped liquid may remain in the casing and seal chamber and may be hot or under pressure. Slacken joints gradually and stand clear of the parting line.

2.2 Personal Protective Equipment

As a minimum, wear safety glasses with side shields, chemical- resistant gloves rated for the pumped fluid, and heat-resistant gloves when the pump may be hot. Face shield and apron are required when breaking joints on hot organic service. Consult the product safety data sheet for fluid-specific requirements.

2.3 Fire and Emergency

Hot organic liquids may be above their flash point. In the event of a seal or joint release, stop the pump, isolate at the nearest accessible valves, and follow the site emergency procedure. Do not attempt to tighten joints on a pump under pressure and at temperature.

2.4 Noise

Sound pressure levels near the pump and driver can exceed 85 dB(A). Hearing protection is required in designated areas. Abnormal noise (cavitation crackle, bearing rumble) is also a diagnostic signal and must be reported, not merely tolerated.

3. Installation

Install the pump on a rigid, level foundation. Grout the baseplate fully to avoid a soft-foot condition, which is a common source of vibration and premature seal and bearing failure. Check that no more than 0.05 mm soft foot exists at each mounting foot.

Pipework must be independently supported so that no piping loads are transmitted to the pump nozzles. Excessive nozzle loads distort the casing and cause internal rubbing and seal misalignment.

3.1 Coupling Alignment

Align the pump and motor shafts by reverse-dial or laser method to within 0.05 mm parallel and 0.05 mm/100 mm angular, cold, with the coupling spacer removed. Re-check alignment hot after the first run, as thermal growth on hot organic duty is significant.

3.2 Seal Flush Piping

The standard seal arrangement is an API Plan 11 flush: a line from the pump discharge through a flow-control orifice to the seal chamber, ensuring positive flush flow across the seal faces and venting of vapour. The flush orifice size is stamped on the seal cartridge and listed in Section 7. Do NOT substitute a different flush plan (for example a bypass "Plan Y" arrangement) without OEM approval - inadequate flush is the leading cause of seal-face distress and leakage on this pump. Verify the installed flush plan against this manual at commissioning and after any seal work.

3.3 Baseplate and Grouting

The baseplate must be fully grouted with a non-shrink grout, with all voids filled, to provide a rigid, continuous support. A partially grouted or hollow baseplate behaves as a soft foot and is a persistent source of vibration. After grouting has cured, re-check that each foot sits flat with no more than 0.05 mm gap before final bolt-up.

3.4 Suction Piping and NPSH

Design the suction line short and direct, with a steady rise to the pump and no high points that can trap vapour. Provide the NPSH margin stated on the data sheet; hot organic liquids are often near their bubble point and are unforgiving of suction losses. A fouled suction strainer or an undersized suction line will induce cavitation, unstable flow and impeller erosion.

A temporary conical suction strainer is recommended for commissioning to protect the pump from construction debris. Remove or replace it once the system is clean; a neglected strainer is a common cause of gradual flow loss.

4. Operation

Prime the pump and open the seal flush before starting. Start against a nearly closed discharge and open the discharge valve promptly to reach the rated operating region. Do not run below the minimum continuous stable flow stamped on the nameplate.

Monitor bearing housing temperature, seal chamber temperature, flush flow and vibration during operation. A rising seal chamber temperature or falling flush flow indicates a flush restriction and must be investigated before the seal is damaged.

4.1 Normal Operating Limits

Bearing housing temperature should not exceed 85 C. Overall vibration should remain below 4.5 mm/s RMS. Seal flush flow must remain within the range on the seal data sheet at all times.

4.2 Start-up

Confirm the pump is primed and the suction valve is fully open. Open the seal flush and confirm flush flow before starting. Start the driver against a nearly closed discharge valve, then open the discharge promptly to bring the pump into its operating region. Do not run against a closed discharge for more than a few seconds as the liquid will overheat and flash.

4.3 Shutdown

Close the discharge valve slowly and stop the driver. Keep the seal flush in service until the seal chamber has cooled. For hot organic service, allow the pump to cool gradually before draining to avoid thermal shock and to protect the seal faces.

4.4 Routine Monitoring

On each operator round, record bearing housing temperature, seal chamber temperature, flush flow indication, discharge pressure and overall vibration. Trend these readings; a slow rise in seal chamber temperature or a fall in flush flow is an early indicator of a developing flush restriction and should be actioned before the seal is damaged. Report any external leakage at the gland immediately.

4.5 Minimum Flow and Parallel Operation

Do not operate continuously below the minimum continuous stable flow on the data sheet; low-flow operation raises temperature, radial load and internal recirculation. When two sisters run in parallel, confirm both are sharing load and neither is running back on its curve near shut-off.

5. Maintenance

Maintenance intervals are given in Section 5.1. The single most important maintenance item on hot organic duty is the mechanical seal and its flush system. The cartridge seal replacement procedure is given in full in Section 5.3.

5.1 Maintenance Schedule

Weekly: check bearing and seal chamber temperature, vibration, flush flow and for any external leakage. Monthly: check coupling condition and alignment indication. Annually: inspect bearings, renew lubricant, inspect the seal and verify the flush orifice and flush plan against this manual.

5.2 Bearing Lubrication

Bearings are grease-lubricated as standard. Use only the specified grade. Over-greasing causes overheating; contamination or water ingress degrades the lubricant and accelerates bearing wear.

5.3 Seal Cartridge Replacement Procedure

The MSC-CP200 uses a self-contained cartridge mechanical seal to make replacement fast and repeatable. Follow the numbered steps below. Cleanliness of the seal faces and correct flush restoration are critical to seal life.

1. Isolate, de-pressurise, drain and cool the pump; apply LOTO and confirm zero energy (see Section 2 and the site LOTO SOP).
2. Disconnect and remove the coupling spacer; do not disturb the alignment shims.
3. Disconnect the seal flush piping at the seal chamber and cap the lines to keep them clean.
4. Remove the bearing housing / back pull-out assembly from the casing, leaving the casing and pipework in place.
5. Release the cartridge clamping ring set-screws and withdraw the spent cartridge seal from the shaft as a complete unit.
6. Clean the shaft and seal chamber faces; inspect the shaft sleeve seat for scoring or fretting and renew if damaged.
7. Fit the new cartridge seal, ensuring the setting clips are in place; do NOT remove the clips until the assembly is complete.
8. Reassemble the back pull-out into the casing; torque the casing bolts evenly to the values in Section 7.
9. Tighten the cartridge drive collar, then remove and retain the seal setting clips.
10. Reconnect the seal flush piping. Verify the flush plan is OEM Plan 11 and the flush orifice is the correct size - do not leave a substitute flush arrangement in service.
11. Refit the coupling spacer and re-check alignment; correct if disturbed.
12. Remove LOTO, restore the flush, prime, and start the pump. Confirm positive flush flow and no leakage before returning to service.

5.4 Bearing Replacement

Renew bearings at the scheduled interval or when condition monitoring indicates wear (rising vibration, noise or bearing housing temperature). Cleanliness is essential; a single hard particle will destroy a new bearing.

1. Isolate, drain, cool and apply LOTO as for seal work (Section 5.3, steps 1-2).
2. Remove the back pull-out assembly and, if fitted, the cartridge seal.
3. Record the existing shim pack; remove the bearing end cover and lock-nut.
4. Withdraw the shaft from the bearing housing and remove the old bearings with a proper puller - never hammer.
5. Clean the shaft and housing; inspect the shaft seats and housing bore for damage.
6. Heat the new bearings inductively to the recommended temperature and fit them square to the shoulder; do not exceed the maximum bearing temperature.
7. Reassemble with the recorded shim pack; set and check the end float.
8. Re-lubricate with the correct grade and quantity; do not over-grease.
9. Reassemble the back pull-out and return the pump to service per Section 5.3.

5.5 Seal Flush System Maintenance

The API Plan 11 flush is the seal's lifeline on hot organic duty and is the single most important item to keep in specification. The flush takes a controlled bleed from the pump discharge, through a flow orifice, into the seal chamber, cooling and lubricating the faces and sweeping away vapour.

At every seal inspection, confirm the flush line is clear, the orifice is the correct size and not eroded or plugged, and any flush strainer is clean. Confirm the flush arrangement is the OEM-specified Plan 11 and has not been altered in the field to a non-standard "Plan Y" bypass. A restricted or non-standard flush starves the faces, drives them hot, and produces the scored faces and gland weeping that precede a full seal failure.

If repeated seal failures occur across sister pumps in the same service, treat the flush system as the prime suspect and audit the flush plan on every sister against this manual before renewing further seals. Renewing a seal without correcting an inadequate flush simply repeats the failure.

5.6 Fault Diagnosis Approach

Diagnose from evidence, not assumption. Distinguish leakage faults (seal, flush, gasket) from vibration faults (misalignment, cavitation, bearing, soft foot) using the symptom, the flush and temperature trends, and vibration data. The troubleshooting table in Section 6 maps common symptoms to their probable causes and the correct first action.

6. Troubleshooting

The table below lists common symptoms, their probable causes and the recommended action. For seal leakage, always verify the seal flush plan and flush line against Section 3.2 before renewing the seal - renewing a seal without correcting an inadequate flush will only repeat the failure.

Symptom	Probable cause	Recommended action
Seal leakage; product weeping or dripping at gland	Inadequate seal flush - flush line blocked or non-OEM flush plan (e.g. Plan Y); worn or scored faces	Verify seal flush plan against OEM Plan 11 (Section 3.2); inspect and clear flush line/orifice/strainer; renew cartridge seal if faces damaged
Seal chamber temperature rising; flush flow low	Flush line restriction, fouled orifice or blocked strainer	Inspect and clean the flush line and orifice; confirm flush flow to seal data sheet
High vibration	Shaft misalignment, cavitation, or soft foot	Check coupling alignment (laser) and correct; verify NPSH margin against cavitation; check hold-down bolts / soft foot
Crackling / gravel noise, unstable flow	Cavitation from insufficient NPSH or suction restriction	Restore suction conditions; clean suction strainer; confirm NPSH available exceeds required
Bearing noise / high bearing temperature	Bearing wear, lubrication degradation or over-greasing	Renew bearings; use correct lubricant grade and quantity
Low developed head / flow	Impeller wear or damage, or worn wear rings	Inspect and renew impeller and wear rings; reset clearances
Loss of drive / sudden vibration	Coupling element failure	Renew coupling element and bolts; re-check alignment

7. Spare Parts

The recommended spares for the MSC-CP200 are listed below. Always quote the pump serial number and the part number when ordering. The cartridge seal and the flush orifice are the critical spares for hot organic duty and should be held in stock for each installed pump.

7.1 Recommended Spare Parts

Item	Part number	Material	Qty
Cartridge mechanical seal	RW-CP200-SEAL-C	SiC/Carbon, Viton	1
Seal flush orifice (Plan 11)	RW-CP200-ORF-3.0	SS316	1
Impeller	RW-CP200-IMP	12% Cr	1
Wear ring set	RW-CP200-WR	12% Cr	1 set
Drive-end bearing	RW-CP200-BRG-DE	-	1
Non-drive-end bearing	RW-CP200-BRG-NDE	-	1
Shaft sleeve	RW-CP200-SLV	SS316	1
Casing gasket set	RW-CP200-GKT	Graphite	1 set
Coupling element	RW-CP200-CPL-EL	Elastomer	1
Bearing housing oil seal	RW-CP200-OS	Viton	2
Casing wear ring	RW-CP200-CWR	12% Cr	1
Impeller wear ring	RW-CP200-IWR	12% Cr	1
Shaft key	RW-CP200-KEY	SS316	1
Bearing lock-nut and washer	RW-CP200-LN	Steel	1 set
Seal chamber gasket	RW-CP200-SCG	Graphite	1
Deflector	RW-CP200-DEF	NBR	1
Casing drain plug	RW-CP200-DP	SS316	1
Flush line tubing kit	RW-CP200-FLK	SS316	1 set

7.2 Fastener Torque Values

Fastener	Torque
Casing bolts M16	210 Nm
Bearing housing bolts M12	85 Nm
Coupling hub set-screw	35 Nm
Cartridge drive collar	12 Nm

8. Technical Data

Nominal technical data for the MSC-CP200 frame is given below. The duty-specific values are on the nameplate and the pump data sheet.

Parameter	Value
Configuration	Single-stage overhung, OH2
Max flow	220 m3/h
Max differential head	160 m
Max working temperature	220 C
Max working pressure	25 bar(g)
Seal type	Cartridge mechanical, API Plan 11
Bearing lubrication	Grease (standard)
Bearing temp alarm	85 C
Vibration alarm	4.5 mm/s RMS
Coupling	Spacer, flexible element

A. Appendix - Installed Pump Data Sheets

A.1 P-3401 - Ester feed pump A

Parameter	Value
Tag	P-3401
Service duty	Hot organics transfer - molten ester feed, 185 C
Plant / Unit	MT / U300
Area	Esterification
Criticality	A
Installed on	2019-05-14
Seal flush plan	API Plan 11 (OEM)

A.2 P-3402 - Ester feed pump B

Parameter	Value
Tag	P-3402
Service duty	Hot organics transfer - ester product, 175 C
Plant / Unit	MT / U300
Area	Product Transfer
Criticality	A
Installed on	2019-05-14
Seal flush plan	API Plan 11 (OEM)

A.3 P-3105 - Solvent bottoms pump

Parameter	Value
Tag	P-3105
Service duty	Hot organics transfer - column bottoms, 190 C
Plant / Unit	MV / U100
Area	Distillation
Criticality	A
Installed on	2020-02-20
Seal flush plan	API Plan 11 (OEM)

A.4 P-2210 - Resin transfer pump

Parameter	Value
Tag	P-2210
Service duty	Hot organics transfer - hot resin blend, 160 C
Plant / Unit	MV / U200
Area	Blending
Criticality	B
Installed on	2020-08-03
Seal flush plan	API Plan 11 (OEM)

B. Appendix - Commissioning Checklist

1. Foundation grouted and level; soft foot within 0.05 mm at each foot.
2. Pipework independently supported; no nozzle loads on the pump.
3. Coupling aligned cold to 0.05 mm parallel / angular.
4. Seal flush confirmed as OEM API Plan 11 with correct orifice size.
5. Seal flush line clean and unrestricted; flush flow within data-sheet range.
6. Direction of rotation verified before coupling connected.
7. Bearing lubrication confirmed correct grade and quantity.
8. Suction strainer fitted and clean.
9. Instrumentation (vibration, bearing/seal temperature) commissioned.
10. Hot alignment re-checked after first run at temperature.